

2009
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–10
- All questions are of equal value

Total marks – 120

Attempt Questions 1–10

All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use the Question 1 Writing Booklet.

- (a) Sketch the graph of $y - 2x = 3$, showing the intercepts on both axes. **2**
- (b) Solve $\frac{5x - 4}{x} = 2$. **2**
- (c) Solve $|x + 1| = 5$. **2**
- (d) Find the gradient of the tangent to the curve $y = x^4 - 3x$ at the point $(1, -2)$. **2**
- (e) Find the exact value of θ such that $2 \cos \theta = 1$, where $0 \leq \theta \leq \frac{\pi}{2}$. **2**
- (f) Solve the equation $\ln x = 2$. Give your answer correct to four decimal places. **2**

Question 2 (12 marks) Use the Question 2 Writing Booklet.

(a) Differentiate with respect to x :

(i) $x \sin x$ **2**

(ii) $(e^x + 1)^2$. **2**

(b) (i) Find $\int 5 dx$. **1**

(ii) Find $\int \frac{3}{(x-6)^2} dx$. **2**

(iii) Evaluate $\int_1^4 x^2 + \sqrt{x} dx$. **3**

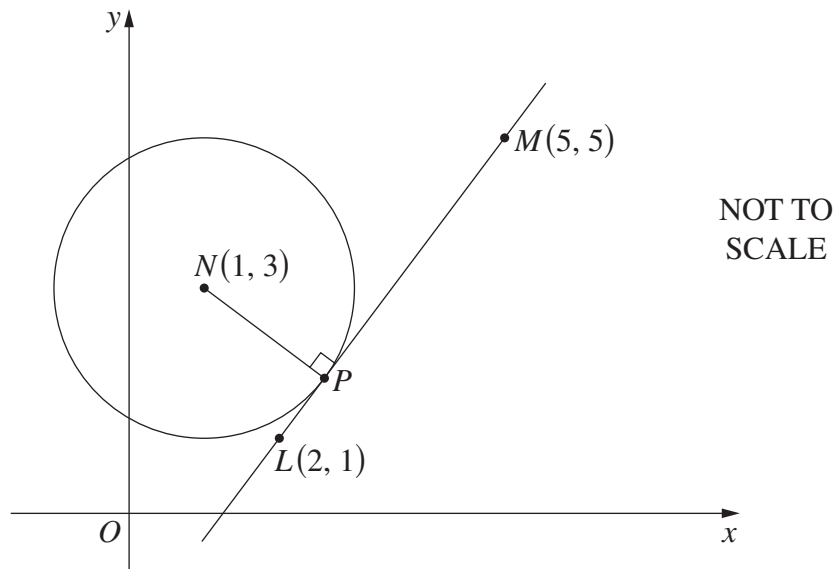
(c) Evaluate $\sum_{k=1}^4 (-1)^k k^2$. **2**

Question 3 (12 marks) Use the Question 3 Writing Booklet.

- (a) An arithmetic series has 21 terms. The first term is 3 and the last term is 53. 2

Find the sum of the series.

(b)



The circle in the diagram has centre N . The line LM is tangent to the circle at P .

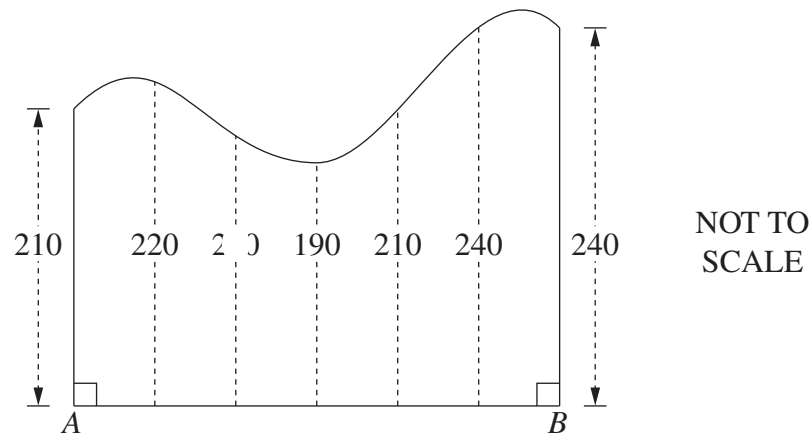
- (i) Find the equation of LM in the form $ax + by + c = 0$. 2
- (ii) Find the distance NP . 2
- (iii) Find the equation of the circle. 1
- (c) Shade the region in the plane defined by $y \geq 0$ and $y \leq 4 - x^2$. 2

Question 3 continues on page 5

Question 3 (continued)

- (d) The diagram shows a block of land and its dimensions, in metres. The block of land is bounded on one side by a river. Measurements are taken perpendicular to the line AB , from AB to the river, at equal intervals of 50 m.

3



Use Simpson's rule with six subintervals to find an approximation to the area of the block of land.

End of Question 3

Question 4 (12 marks) Use the Question 4 Writing Booklet.

- (a) A tree grows from ground level to a height of 1.2 metres in one year. In each subsequent year, it grows $\frac{9}{10}$ as much as it did in the previous year. 2

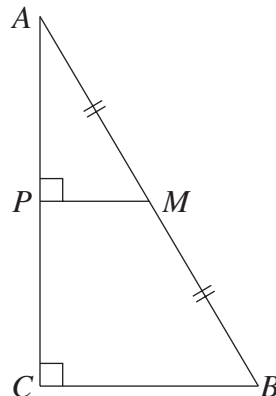
Find the limiting height of the tree.

- (b) Find the values of k for which the quadratic equation 3

$$x^2 - (k + 4)x + (k + 7) = 0$$

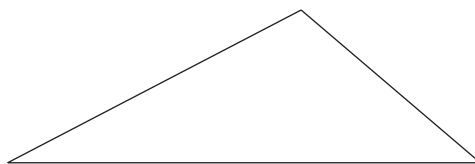
has equal roots.

- (c) In the diagram, $\triangle ABC$ is a right-angled triangle, with the right angle at C . The midpoint of AB is M , and $MP \perp AC$.



Copy or trace the diagram into your writing booklet.

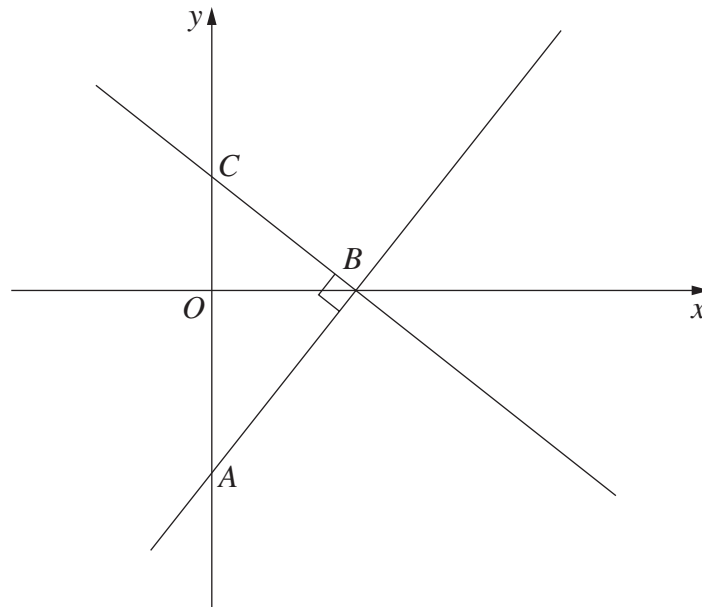
- (i) Prove that $\triangle AMP$ is similar to $\triangle ABC$. 2
- (ii) What is the ratio of AP to AC ? 1
- (iii) Prove that $\triangle AMC$ is isosceles. 2
- (iv) Show that $\triangle ABC$ can be divided into two isosceles triangles. 1
- (v) 1



Copy or trace this triangle into your writing booklet and show how to divide it into four isosceles triangles.

Question 5 (12 marks) Use the Question 5 Writing Booklet.

- (a) In the diagram, the points A and C lie on the y -axis and the point B lies on the x -axis. The line AB has equation $y = \sqrt{3}x - 3$. The line BC is perpendicular to AB .

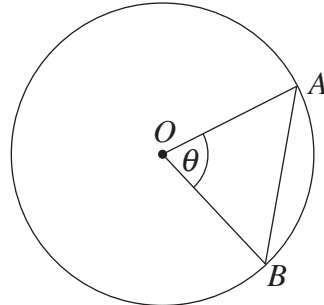


- (i) Find the equation of the line BC . **2**
- (ii) Find the area of the triangle ABC . **2**
- (b) On each working day James parks his car in a parking station which has three levels. He parks his car on a randomly chosen level. He always forgets where he has parked, so when he leaves work he chooses a level at random and searches for his car. If his car is not on that level, he chooses a different level and continues in this way until he finds his car.
- (i) What is the probability that his car is on the first level he searches? **1**
- (ii) What is the probability that he must search all three levels before he finds his car? **1**
- (iii) What is the probability that on every one of the five working days in a week, his car is not on the first level he searches? **1**

Question 5 continues on page 8

Question 5 (continued)

- (c) The diagram shows a circle with centre O and radius 2 centimetres. The points A and B lie on the circumference of the circle and $\angle AOB = \theta$.



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SCALE

- (i) There are two possible values of θ for which the area of $\triangle AOB$ is $\sqrt{3}$ square centimetres. One value is $\frac{\pi}{3}$. **2**

Find the other value.

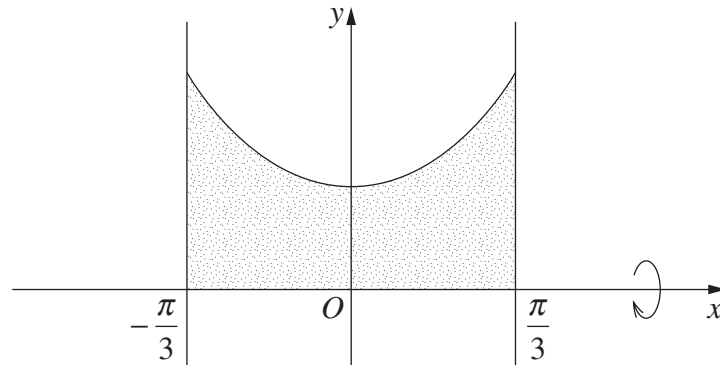
- (ii) Suppose that $\theta = \frac{\pi}{3}$.

- (1) Find the area of the sector AOB . **1**
- (2) Find the exact length of the perimeter of the minor segment bounded by the chord AB and the arc AB . **2**

End of Question 5

Question 6 (12 marks) Use the Question 6 Writing Booklet.

- (a) The diagram shows the region bounded by the curve $y = \sec x$, the lines $x = \frac{\pi}{3}$ and $x = -\frac{\pi}{3}$, and the x -axis. **3**



The region is rotated about the x -axis. Find the volume of the solid of revolution formed.

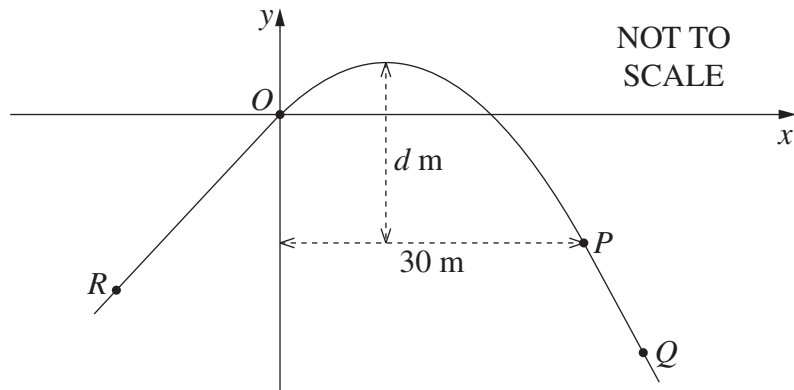
- (b) Radium decays at a rate proportional to the amount of radium present. That is, if $Q(t)$ is the amount of radium present at time t , then $Q = Ae^{-kt}$, where k is a positive constant and A is the amount present at $t = 0$. It takes 1600 years for an amount of radium to reduce by half. **2**
- (i) Find the value of k . **2**
- (ii) A factory site is contaminated with radium. The amount of radium on the site is currently three times the safe level. **2**

How many years will it be before the amount of radium reaches the safe level?

Question 6 continues on page 10

Question 6 (continued)

- (c) The diagram illustrates the design for part of a roller-coaster track. The section RO is a straight line with slope 1.2, and the section PQ is a straight line with slope -1.8 . The section OP is a parabola $y = ax^2 + bx$. The horizontal distance from the y -axis to P is 30 m.



In order that the ride is smooth, the straight line sections must be tangent to the parabola at O and at P .

- | | | |
|------|---|----------|
| (i) | Find the values of a and b so that the ride is smooth. | 3 |
| (ii) | Find the distance d , from the vertex of the parabola to the horizontal line through P , as shown on the diagram. | 2 |

End of Question 6

Question 7 (12 marks) Use the Question 7 Writing Booklet.

- (a) The acceleration of a particle is given by

$$\ddot{x} = 8e^{-2t} + 3e^{-t},$$

where x is displacement in metres and t is time in seconds.

Initially its velocity is -6 m s^{-1} and its displacement is 5 m.

- (i) Show that the displacement of the particle is given by **2**

$$x = 2e^{-2t} + 3e^{-t} + t.$$

- (ii) Find the time when the particle comes to rest. **3**

- (iii) Find the displacement when the particle comes to rest. **1**

- (b) Between 5 am and 5 pm on 3 March 2009, the height, h , of the tide in a harbour was given by

$$h = 1 + 0.7 \sin \frac{\pi}{6} t \text{ for } 0 \leq t \leq 12,$$

where h is in metres and t is in hours, with $t = 0$ at 5 am.

- (i) What is the period of the function h ? **1**

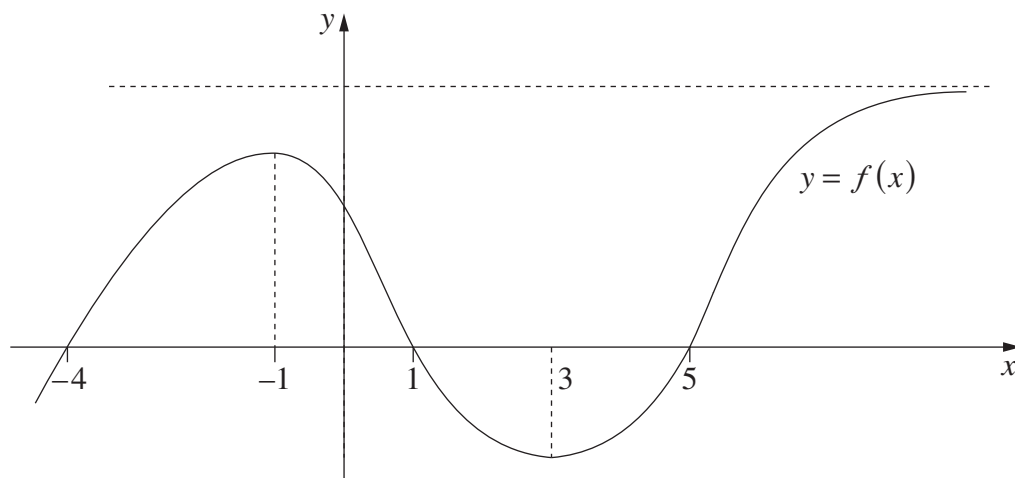
- (ii) What was the value of h at low tide, and at what time did low tide occur? **2**

- (iii) A ship is able to enter the harbour only if the height of the tide is at least 1.35 m. **3**

Find all times between 5 am and 5 pm on 3 March 2009 during which the ship was able to enter the harbour.

Question 8 (12 marks) Use the Question 8 Writing Booklet.

(a)



The diagram shows the graph of a function $y = f(x)$.

- | | | |
|-------|--|----------|
| (i) | For which values of x is the derivative, $f'(x)$, negative? | 1 |
| (ii) | What happens to $f'(x)$ for large values of x ? | 1 |
| (iii) | Sketch the graph $y = f'(x)$. | 2 |

Question 8 continues on page 13

Question 8 (continued)

- (b) One year ago Daniel borrowed \$350 000 to buy a house. The interest rate was 9% per annum, compounded monthly. He agreed to repay the loan in 25 years with equal monthly repayments of \$2937.

(i) Calculate how much Daniel owed after his first monthly repayment. **1**

(ii) Daniel has just made his 12th monthly repayment. He now owes \$346 095. The interest rate now decreases to 6% per annum, compounded monthly. **3**

The amount, $\$A_n$, owing on the loan after the n th monthly repayment is now calculated using the formula

$$A_n = 346\,095 \times 1.005^n - 1.005^{n-1}M - \dots - 1.005M - M$$

where $\$M$ is the monthly repayment, and $n = 1, 2, \dots, 288$. (Do NOT prove this formula.)

Calculate the monthly repayment if the loan is to be repaid over the remaining 24 years (288 months).

(iii) Daniel chooses to keep his monthly repayments at \$2937. Use the formula in part (ii) to calculate how long it will take him to repay the \$346 095. **3**

(iv) How much will Daniel save over the term of the loan by keeping his monthly repayments at \$2937, rather than reducing his repayments to the amount calculated in part (ii)? **1**

End of Question 8

Question 9 (12 marks) Use the Question 9 Writing Booklet.

- (a) Each week Van and Marie take part in a raffle at their respective workplaces. 2

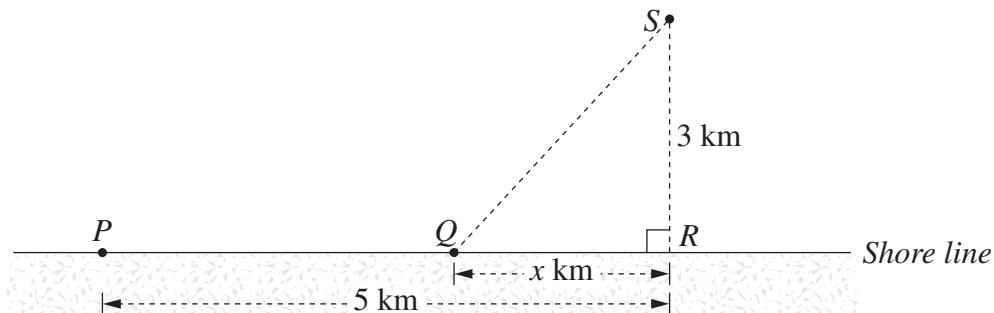
The probability that Van wins a prize in his raffle is $\frac{1}{9}$. The probability that Marie wins a prize in her raffle is $\frac{1}{16}$.

What is the probability that, during the next three weeks, at least one of them wins a prize?

- (b) An oil rig, S , is 3 km offshore. A power station, P , is on the shore. A cable is to be laid from P to S . It costs \$1000 per kilometre to lay the cable along the shore and \$2600 per kilometre to lay the cable underwater from the shore to S .

The point R is the point on the shore closest to S , and the distance PR is 5 km.

The point Q is on the shore, at a distance of x km from R , as shown in the diagram.



- (i) Find the total cost of laying the cable in a straight line from P to R and then in a straight line from R to S . 1
- (ii) Find the cost of laying the cable in a straight line from P to S . 1
- (iii) Let $\$C$ be the total cost of laying the cable in a straight line from P to Q , and then in a straight line from Q to S . 2
- Show that $C = 1000\left(5 - x + 2.6\sqrt{x^2 + 9}\right)$.
- (iv) Find the minimum cost of laying the cable. 4
- (v) New technology means that the cost of laying the cable underwater can be reduced to \$1100 per kilometre. 2

Determine the path for laying the cable in order to minimise the cost in this case.

Question 10 (12 marks) Use the Question 10 Writing Booklet.

Let $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$.

(a) Show that the graph of $y = f(x)$ has no turning points. **2**

(b) Find the point of inflexion of $y = f(x)$. **1**

(c) (i) Show that $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}$ for $x \neq -1$. **1**

(ii) Let $g(x) = \ln(1+x)$. **2**

Use the result in part (c) (i) to show that $f'(x) \geq g'(x)$ for all $x \geq 0$.

(d) On the same set of axes, sketch the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$. **2**

(e) Show that $\frac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$. **2**

(f) Find the area enclosed by the graphs of $y = f(x)$ and $y = g(x)$, and the straight line $x = 1$. **2**

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2009 HSC Mathematics Marking Guidelines

Question 1 (a)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct graph, showing intercepts	2
• Draws a straight line with positive slope, or equivalent merit	1

Question 1 (b)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Makes some progress	1

Question 1 (c)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Obtains one solution	1

Question 1 (d)

Outcomes assessed: P6

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Differentiates correctly, or equivalent merit	1

Question 1 (e)

Outcomes assessed: P4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Makes some progress	1

Question 1 (f)

Outcomes assessed: H3

MARKING GUIDELINES

Criteria	Marks
• Correct answer, rounded correctly	2
• Makes significant progress	1

Question 2 (a) (i)

Outcomes assessed: P7, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Shows understanding of the product rule, or equivalent merit	1

Question 2 (a) (ii)

Outcomes assessed: P7, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Shows understanding of the chain rule, or equivalent merit	1

Question 2 (b) (i)

Outcomes assessed: P8, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 2 (b) (ii)

Outcomes assessed: P8, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Shows some understanding of integrating with negative powers	1

Question 2 (b) (iii)*Outcomes assessed: P8, H5***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Makes some progress	1

Question 2 (c)*Outcomes assessed: H5, H9***MARKING GUIDELINES**

Criteria	Marks
• Correct answer	2
• Shows some understanding of sigma notation	1

Question 3 (a)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Shows some understanding of summing an arithmetic series	1

Question 3 (b) (i)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct equation	2
• Finds correct gradient, or equivalent merit	1

Question 3 (b) (ii)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Attempts to use formula for the distance, or equivalent merit	1

Question 3 (b) (iii)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 3 (c)*Outcomes assessed: P4, P5***MARKING GUIDELINES**

Criteria	Marks
• Correct sketch	2
• Shows some understanding of regions in the plane	1

Question 3 (d)*Outcomes assessed: H8***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	3
• Shows a good understanding of Simpson's rule	2
• Shows some understanding of Simpson's rule	1

Question 4 (a)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Shows some understanding of the sum of a geometric series	1

Question 4 (b)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Obtains $k^2 + 4k - 12 = 0$, or equivalent progress	2
• Attempts to use the discriminant = 0	1

Question 4 (c) (i)

Outcomes assessed: P2

MARKING GUIDELINES

Criteria	Marks
• Proves the triangles similar (showing two pairs of angles equal is sufficient)	2
• Shows some understanding of similar triangles	1

Question 4 (c) (ii)

Outcomes assessed: P2

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 4 (c) (iii)

Outcomes assessed: P2

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Attempts to show $\triangle AMP \equiv \triangle CMP$, or equivalent progress	1

Question 4 (c) (iv)

Outcomes assessed: H2

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 4 (c) (v)

Outcomes assessed: H2

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 5 (a) (i)

Outcomes assessed: P4

MARKING GUIDELINES

Criteria	Marks
• Correct equation	2
• Finds correct gradient, or equivalent merit	1

Question 5 (a) (ii)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Finds relevant lengths, or equivalent progress	1

Question 5 (b) (i)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (b) (ii)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (b) (iii)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (c) (i)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Attempts to use the formula for area, or equivalent merit	

Question 5 (c) (ii) (1)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (c) (ii) (2)

Outcomes assessed: P2, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Finds length of chord or length of arc, or equivalent merit	1

Question 6 (a)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Correct primitive, or equivalent progress	2
• Shows some understanding of how to compute volumes of revolution	1

Question 6 (b) (i)

Outcomes assessed: H3, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Obtains $e^{-1600k} = \frac{1}{2}$, or equivalent merit	1

Question 6 (b) (ii)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Makes some progress	1

Question 6 (c) (i)

Outcomes assessed: P2, P3, P6

MARKING GUIDELINES

Criteria	Marks
• Solves to find the values of a and b	3
• Solves to find the value of a , or of b , or equivalent merit	2
• Matches the derivatives at O , or at P , or equivalent merit	1

Question 6 (c) (ii)

Outcomes assessed: P3, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Finds the y -coordinate at P , or finds the y -coordinate at the vertex, or equivalent merit	1

Question 7 (a) (i)

Outcomes assessed: H5, H9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Shows an understanding of the need to integrate or differentiate twice	1

Question 7 (a) (ii)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	3
• Solves the correct quadratic, or equivalent merit	2
• Attempts to solve $\dot{x} = 0$ as a quadratic in e^{-t}	1

Question 7 (a) (iii)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 7 (b) (i)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
Correct answer	1

Question 7 (b) (ii)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
Both correct answers	2
Correct height at low tide, or correct value for t	1

Question 7 (b) (iii)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
Correct time interval — i.e. between 6 am and 10 am	3
Makes significant progress	2
Attempts to solve $h \geq 1.35$	1

Question 8 (a) (i)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 8 (a) (ii)

Outcomes assessed: H6, H7

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 8 (a) (iii)

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
• Correct graph showing all important features	2
• Draws a graph showing some of the important features	1

Question 8 (b) (i)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 8 (b) (ii)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	3
• Makes substantial progress	2
• Obtains an expression for M by setting $A_{288} = 0$	1

Question 8 (b) (iii)

Outcomes assessed: H2, H4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	3
• Makes substantial progress	2
• Obtains the equation $346\,095 \times 1.005^n - 2937 \left(\frac{1.005^n - 1}{0.005} \right) = 0$, or equivalent merit	1

Question 8 (b) (iv)

Outcomes assessed: H4, H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 9 (a)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Shows some understanding of complementary events, or equivalent merit	1

Question 9 (b) (i)

Outcomes assessed: P3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 9 (b) (ii)

Outcomes assessed: P3, P4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 9 (b) (iii)

Outcomes assessed: P3, P4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Correctly identifies the cost for one section, or equivalent merit	1

Question 9 (b) (iv)

Outcomes assessed: H5, H9

MARKING GUIDELINES

Criteria	Marks
• Correct solution, including justification [using parts (i) and (ii) or otherwise]	4
• Obtains $C = 12\,200$	3
• Correctly solves $\frac{dC}{dx} = 0$, or equivalent progress	2
• Correctly differentiates C , or equivalent progress	1

Question 9 (b) (v)

Outcomes assessed: H5, H9

MARKING GUIDELINES

Criteria	Marks
• Correct description of path	2
• Makes some progress	1

Question 10 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Attempts to shows $f'(x) \neq 0$	1

Question 10 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 10 (c) (i)

Outcomes assessed: P3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 10 (c) (ii)

Outcomes assessed: H3, H5, H9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Obtains $g'(x)$, or equivalent progress	1

Question 10 (d)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
Correct sketch showing: correct graphs for $y = g(x)$ and $y = f(x)$, with $f(0) = 0$; $g(0) = 0$, and $f(x) > g(x)$ for $x > 0$	2
Correct sketch showing at least two of the above	1

Question 10 (e)

Outcomes assessed: H3, H5

MARKING GUIDELINES

Criteria	Marks
Correct solution	2
Makes some progress	1

Question 10 (f)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Correct answer	2
Attempts to use the result in part (e), or equivalent merit	1

Mathematics

2009 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (a)	2	6.1	P4
1 (b)	2	1.4	P4
1 (c)	2	1.4, 9.1	P4
1 (d)	2	8.5	P6
1 (e)	2	5.2, 5.3, 13.1	P4, H5
1 (f)	2	12.4	H3
2 (a) (i)	2	8.8, 12.4	P7, H5
2 (a) (ii)	2	8.8, 12.4	P7, H5
2 (b) (i)	1	11.2	P8, H5
2 (b) (ii)	2	11.2	P8, H5
2 (b) (iii)	3	11.1, 11.2	P8, H5
2 (c)	2	7.0	H5, H9
3 (a)	2	7.1	H5
3 (b) (i)	2	6.2	P4
3 (b) (ii)	2	6.5	P4
3 (b) (iii)	1	4.3	P4
3 (c)	2	4.4, 9.1	P4, P5
3 (d)	3	11.3	H8
4 (a)	2	7.3	H4, H5
4 (b)	3	9.2	P4
4 (c) (i)	2	2.3	P2
4 (c) (ii)	1	2.3	P2
4 (c) (iii)	2	2.3	P2
4 (c) (iv)	1	2.5	H2
4 (c) (v)	1	2.5	H2
5 (a) (i)	2	6.2	P4
5 (a) (ii)	2	6.8	H5
5 (b) (i)	1	3.1	H5
5 (b) (ii)	1	3.2	H5
5 (b) (iii)	1	3.2	H5
5 (c) (i)	2	5.5, 13.1	H5
5 (c) (ii) (1)	1	13.1	H5
5 (c) (ii) (2)	2	2.3, 13.1	P2, H5
6 (a)	3	11.4, 13.6	H8

Question	Marks	Content	Syllabus outcomes
6 (b) (i)	2	14.2	H3, H5
6 (b) (ii)	2	14.2	H4, H5
6 (c) (i)	3	8.6, 9.1	P2, P3, P6
6 (c) (ii)	2	9.1, 10.2	P3, H5
7 (a) (i)	2	14.3	H5, H9
7 (a) (ii)	3	14.3	H5
7 (a) (iii)	1	14.3	H5
7 (b) (i)	1	13.7	H4, H5
7 (b) (ii)	2	13.7	H4, H5
7 (b) (iii)	3	13.7	H4, H5
8 (a) (i)	1	10.1	H7
8 (a) (ii)	1	10.2	H6, H7
8 (a) (iii)	2	10.1, 10.2	H7, H9
8 (b) (i)	1	7.5	H4, H5
8 (b) (ii)	3	7.5	H4, H5
8 (b) (iii)	3	7.5	H2, H4, H5
8 (b) (iv)	1	7.5	H4, H5
9 (a)	2	3.1, 3.2, 3.3	H5
9 (b) (i)	1	1.1	P3
9 (b) (ii)	1	1.1, 2.3	P3, P4
9 (b) (iii)	2	1.1, 1.3, 2.3	P3, P4
9 (b) (iv)	4	1.3, 10.6	H5, H9
9 (b) (v)	2	10.6	H5, H9
10 (a)	2	10.2	H6
10 (b)	1	10.2, 10.4	H6
10 (c) (i)	1	1.3	P3
10 (c) (ii)	2	8.8, 10.1, 12.5	H3, H5, H9
10 (d)	2	10.5, 12.5	H6, H9
10 (e)	2	8.8, 12.5	H3, H5
10 (f)	2	11.4	H8